

Predicting Hourly Electricity Demand in Toledo

IOE 562: Statistical Models and Analysis for Data Science

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Abstract

We predict hourly electricity demand for Toledo, Ohio using hourly load data from Toledo Edison and weather data from Open-Meteo. The raw merged dataset covers January 2019 through December 2025 and has 61,227 hours. A binary indicator was implemented on the COVID window (March 1, 2020 through June 1, 2021), and all hours were kept for modeling. We built the forecast in two parallel approaches so we could compare the work of these methods. We then added a deep-learning model as a third comparison point. Approach 1 is a regression pipeline that starts from OLS, prunes collinear variables using VIF, transforms the response via Box-Cox (log), replaces linear temperature with a B-spline, and then adds a lagged demand term. Approach 2 uses the same cleaned feature matrix and runs OLS, Ridge, LASSO, WLS, PCR, Huber, LAD, LTS, SARIMA, and SARIMAX on a shared chronological three-way split. Approach 3 is a two-layer LSTM on the same engineered features.

Huber regression was the best of the linear family at 86,528 KW test RMSE, OLS/Ridge/LASSO/WLS/PCR landed within a few hundred KW of Huber (86,700 to 87,200 KW), SARIMA and SARIMAX were far behind (337,752 and 318,583 KW), and the LSTM produced the lowest test RMSE overall at 78,432 KW with test MAE 62,514 KW. A one-to-one demand-to-bill link is a reasonable first-order assumption (a 20% increase in demand maps to roughly a 20% higher bill before accounting for any demand charges or tiered rates), so forecast error in kilowatts translates directly to dollars for customers. Future work should request the Toledo Edison rate schedule to convert RMSE into dollar error and should consider modeling residential demand separately from commercial and industrial loads.

1. Introduction

Grid operators have to balance supply and demand in real time. If they overestimate demand they waste fuel and money. If they underestimate it they risk brown-outs or forced load shedding. Weather is the single largest short-term driver because households and businesses change their heating and cooling in response to temperature, humidity, and solar load, so a forecast that gets weather right tends to get demand right.

Demand also ties back to customer bills. Most of a typical electricity bill scales with the kilowatt-hours consumed, so a 20% forecast error in aggregate demand maps, to a first approximation, to a 20% swing in the utility's revenue and in individual customer bills (actual rate structures include tiered charges, demand charges, and seasonal riders, so the real mapping is more nuanced and is discussed as future work in Section 8). This makes the forecast relevant to both the utility and the customer, not just to grid dispatch.

We originally planned to use MISO grid data paired with a Detroit weather station. MISO covers fifteen states, so pairing it with a single weather station did not make geographic sense. We switched to Toledo Edison, the investor-owned utility that serves the Toledo, Ohio metro area, and paired the load data with Open-Meteo reanalysis weather for the same city. This gives a tight geographic match between the consumers and the weather they actually experienced.

We built the model in two parallel approaches so each of us could cross-check the two approaches pipelines, and then added a deep-learning model (Approach 3, the LSTM) as an out-of-family reference point. Approach 1 focuses on getting the nonlinear temperature effect right through a B-spline and then adding a one-hour lag of demand. Approach 2 runs a broader suite of estimators (shrinkage, robust, and explicit time-series models) on the same cleaned feature matrix. The LSTM is outside the statistical-regression scope but gives us a sanity check on how much structure the regression family is missing.

2. Data Description and Preparation

2.1 Data Sources

Electricity consumption data comes from Toledo Edison (a FirstEnergy Ohio operating company). The raw Excel file contains hourly load broken down by customer class (Commercial, Industrial, Residential) for both regular and PIPP (Percentage of Income Payment Plan) customers. For each hour we summed Non-Shopped + Shopped + PIPP load to get total consumption in kilowatts. The series spans January 2019 through February 2026 (62,784 hourly rows before merging). We used total load rather than residential-only because the course context is grid-level forecasting; splitting the response into residential / commercial / industrial components is listed as future work.

Weather data comes from Open-Meteo, which provides model-based reanalysis for the Toledo grid cell from January 2019 through December 2025. Reanalysis has one practical advantage over raw NOAA station data: zero missing values. The file contains 19 variables - temperature and apparent temperature (F), humidity, wind speed, wind gusts, wind direction, cloud cover, shortwave radiation, direct radiation, direct radiation instant, soil temperature, soil moisture, vapor-pressure deficit, reference evapotranspiration, precipitation, rain, snowfall, snow depth, and a day/night indicator.

2.2 Data Preparation

The main cleaning issues were the Excel file's mixed datetime formats (some sheets had separate date + hour columns, others had a combined datetime string), the convention that hour 24 means midnight of the next day, and stray thousands separators in numeric strings. Weather data was converted from UTC to Eastern time before merging on the hourly timestamp. The merged dataset has 61,227 rows, 21 base columns, and zero missing values.

We added engineered features consistently across approaches: hour-of-day, day-of-week, month, and season as factors, plus binary flags for weekend, U.S. federal holiday, the 2020-2021 COVID lockdown window, and hot heat-wave days (apparent temperature above 90F). Approach 1 adds a one-hour lag of demand, a 24-hour rolling mean of apparent temperature, and an hour-by-season interaction. Approach 2 adds sine and cosine encodings of hour and day-of-year so that SARIMAX has smooth cyclical regressors.

We decided to keep the COVID window (March 1, 2020 through June 1, 2021) and introduce a binary indicator for COVID before modeling. Lockdown demand is not a function of weather or the calendar - it is a regime shift driven by policy - and the flag allows the model to learn the average demand shift during that window while keeping all 61,227 hours for modeling.

3. Exploratory Data Analysis

3.1 Response Distribution and Time Series

Hourly demand ranges from 710,935 to 2,400,995 KW with mean 1,310,413 KW and median 1,298,487 KW. The distribution is right-skewed: the mode sits near 1.1 million KW and the right tail runs past 2.3 million during summer peaks. The seven-year time series shows strong annual seasonality (summer peaks, spring troughs) and no long-term trend. The 2020-2021 pandemic window is visibly lower in the raw series.

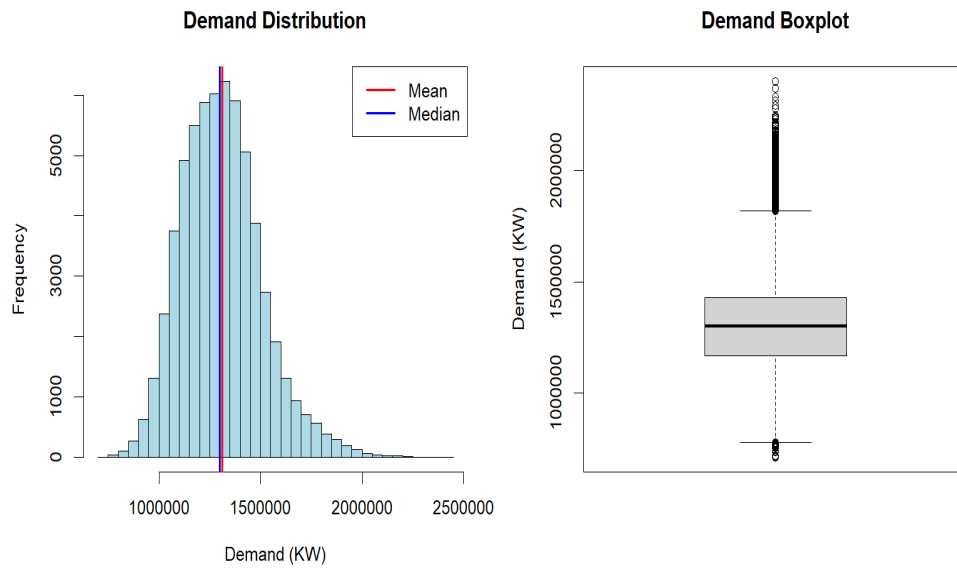


Figure 1: Distribution of hourly demand across 2019-2025. Right-skewed with a long summer tail.

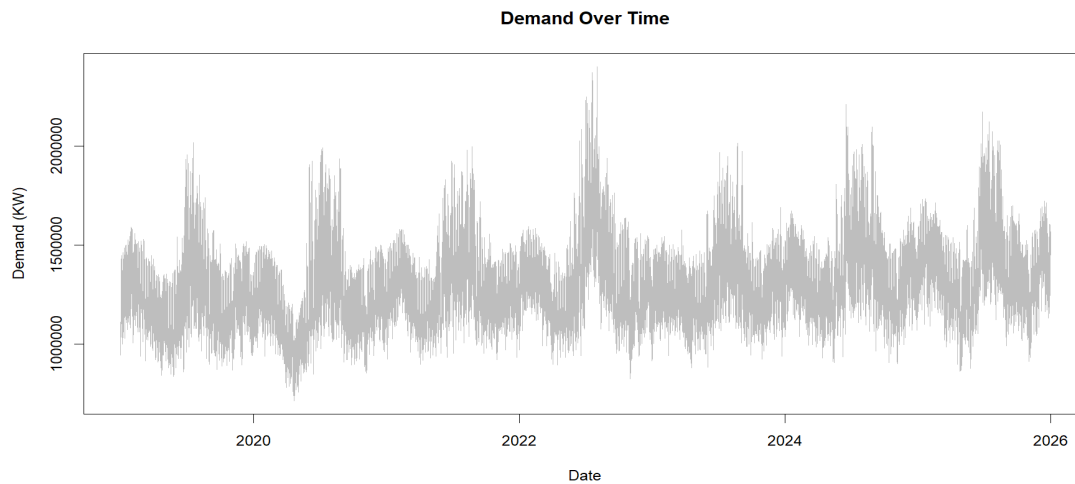


Figure 2: Hourly demand over time. Summer peaks dominate the series.

3.2 Demand versus Weather

Demand versus apparent temperature shows a U-shape: demand is lowest around 50-60F, rises sharply above 70F as air conditioning comes on, and rises more moderately below 30F for heating. The cooling arm is noticeably steeper than the heating arm, which is consistent with the Midwest's natural-gas-dominated heating mix. Because the curve is nonlinear, a plain linear temperature term fits poorly at the edges, so Approach 1 replaces it with a B-spline (knots at 30, 50, and 70F) and Approach 2 tries a cubic polynomial.

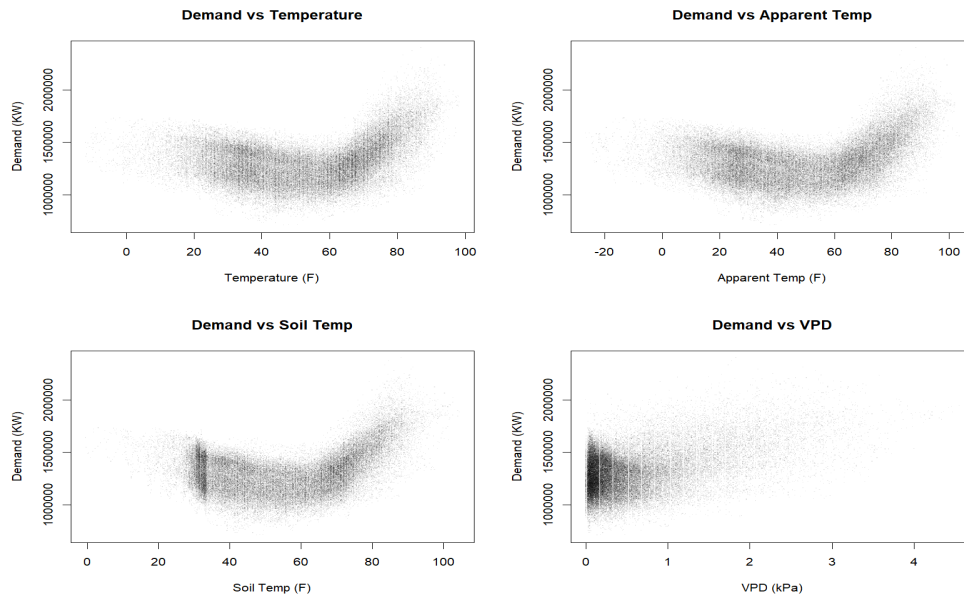


Figure 3: Demand vs apparent temperature, colored by season. The asymmetric U-shape motivates the spline.

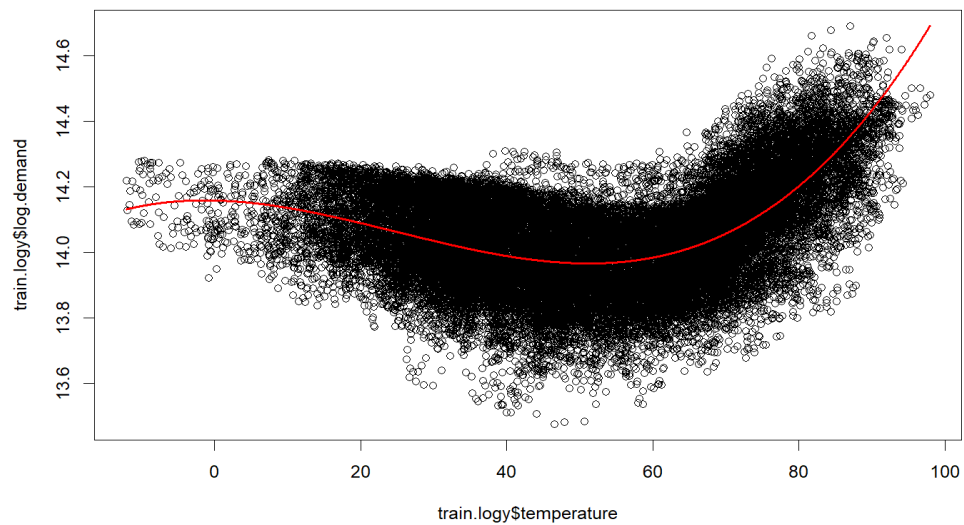


Figure 4: Cubic polynomial fit of demand on apparent temperature used in Approach 2.

3.3 Time Patterns

Demand rises through the morning, peaks between 14:00 and 17:00, and drops to its daily low around 02:00-05:00. Summer months are highest, spring lowest, and weekends run about 5% below weekdays. All three signals are highly significant factors in every model we fit, which is why we treated hour, day-of-week, month, and season as factor variables from the start rather than as continuous proxies.

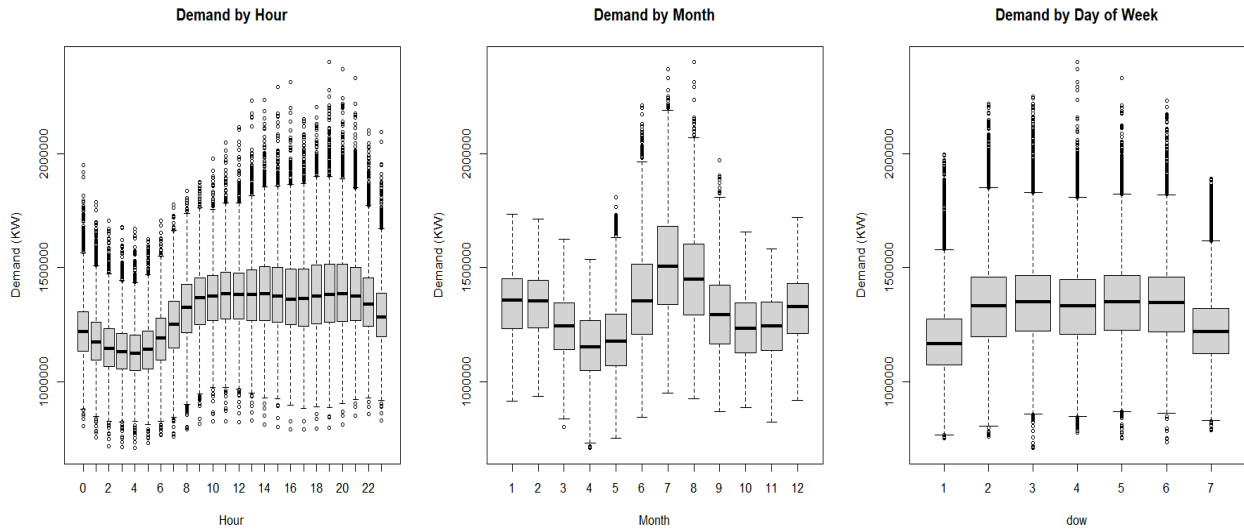


Figure 5: Demand by hour, day-of-week, and month. All three are strong predictors.

3.4 Correlation and Autocorrelation

The correlation heatmap flagged several near-duplicate weather variables. Temperature, apparent temperature, and soil temperature all correlate above 0.97. The radiation variables exceed 0.95 with each other. Precipitation and rain are 0.98. Wind speed and wind gusts are 0.95. VPD has the strongest raw linear correlation with demand at 0.43. These clusters drove the VIF pruning step in Section 4.

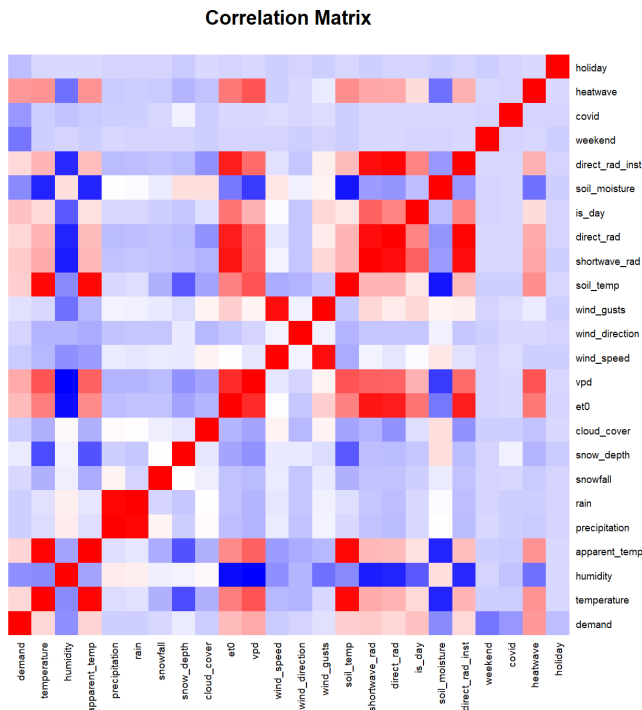


Figure 6: Correlation heatmap of weather predictors. Multiple collinear clusters are visible.

The autocorrelation function of demand decays slowly and has sharp peaks at lags 24, 48, and 168 hours, which confirms a daily and a weekly cycle. This is the structure SARIMA is built for and the reason we eventually added a lag-1 term inside the regression.

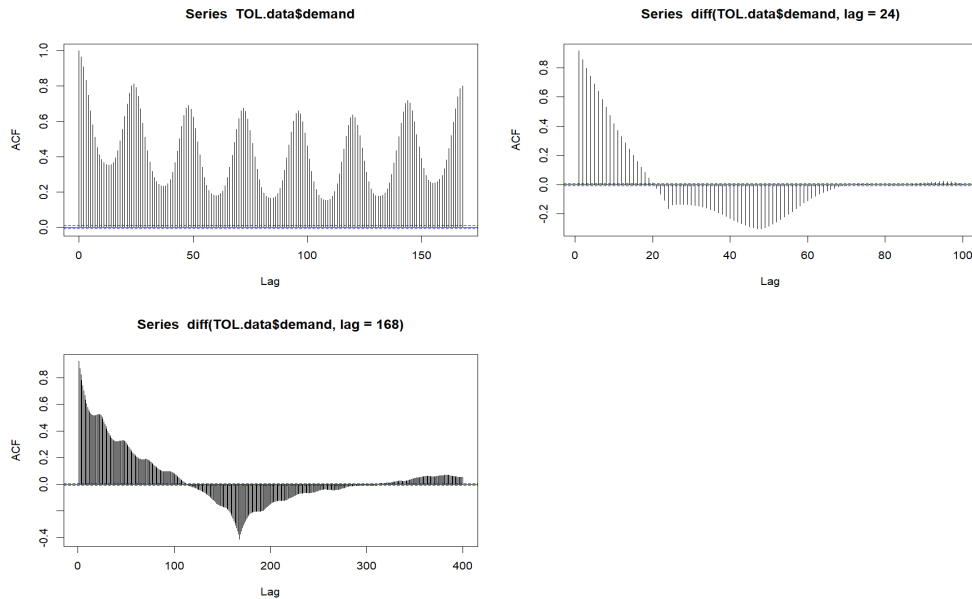


Figure 7: Sample ACF of hourly demand. Peaks at 24 and 168 hours confirm daily and weekly seasonality.

4. Approach 1 - B-spline Regression with Lagged Demand

Approach 1 is a linear-regression pipeline that replaces the linear temperature term with a B-spline, takes a Box-Cox log transform of the response, and then adds a one-hour lag of demand plus an hour-by-season interaction. The word "autoregressive" is avoided here on purpose - adding lag-1 of the response to an OLS design is not the same thing as fitting an AR model, and we did not want the two to be confused. The goal in this approach is to see how far the OLS family can be pushed on this dataset.

4.1 Model Development Path

Starting from an OLS model with all 19 weather predictors and time factors (initial adjusted R-squared 0.743, sigma 103,176 KW), we worked through a standard diagnostic pipeline: variance-inflation-factor screening to drop six collinear weather variables (rain, direct radiation, direct radiation instantaneous, reference ET, wind gusts, soil temperature), BIC selection to drop humidity, precipitation, snowfall, wind direction, and is_day, and a Box-Cox check that gave a 95% confidence interval for lambda of approximately -0.1 (CI [-0.25, 0]); log remained inside the CI boundary and was retained for interpretability. We adopted log(demand) as the response, which visibly compressed the fan shape in the residuals-vs-fitted plot and halved the QQ-plot tail deviations.

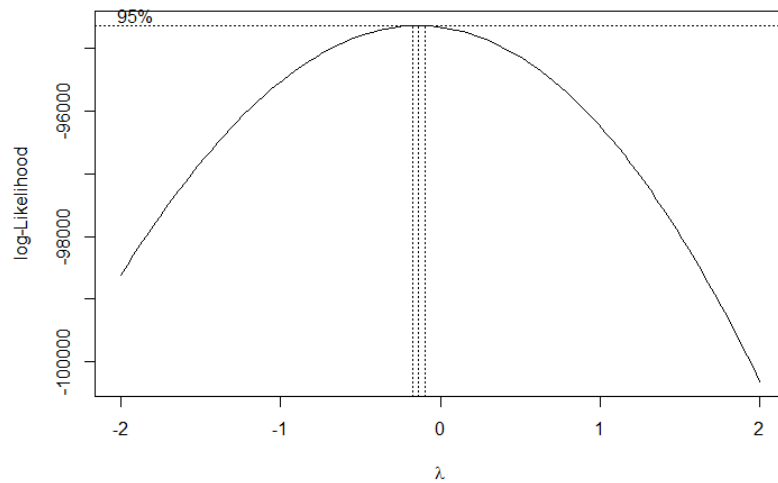


Figure 8: Box-Cox profile for the reduced OLS model. Lambda estimate is approximately -0.1 with CI [-0.25, 0]; log remains at the CI boundary and is retained for interpretability.

We then tested a sequence of 16 candidate specifications (linear, quadratic, cubic, spline, and various interaction terms on the log response). The B-spline with knots at 30, 50, and 70F plus the holiday flag won by a clear margin: test RMSE 120,954 KW, MAPE 7.49%, and adjusted R-squared 0.756, roughly four percentage points of R-squared better than the cubic polynomial. We call this the Primary model.

4.2 Primary Model

It was evaluated on a three-way chronological split (train pre-2023, 35,050 hours; validation Jan–Oct 2023, 7,294 hours; test Nov 2023+, 18,883 hours) matching the split used in Approach 2. The training-set residual scale and the held-out RMSE are consistent, so the held-out number (120,954 KW test RMSE) is not a lucky draw. The held-out MAPE is 7.49% and 95% prediction-interval coverage is 0.923. Residual diagnostics are clean: the fan shape is gone after the log transform, the QQ tail is mild, and Cook's distance stays below 0.005 for every observation (well below the conventional 1.0 threshold).

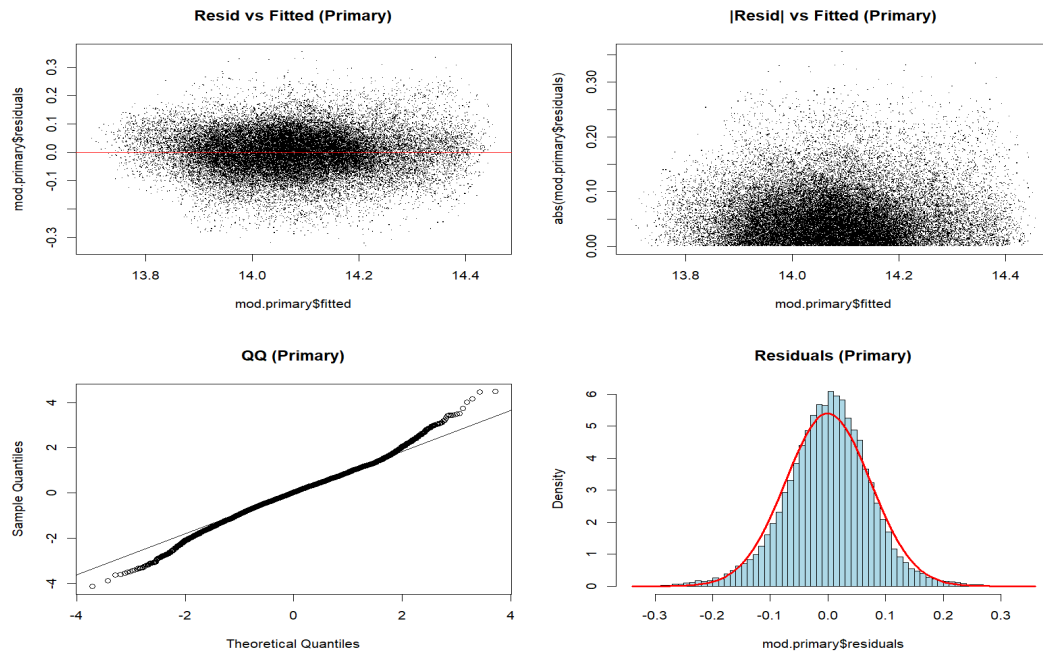


Figure 9: Residual diagnostics for the Primary model. Heteroscedasticity is substantially reduced after the log transform.

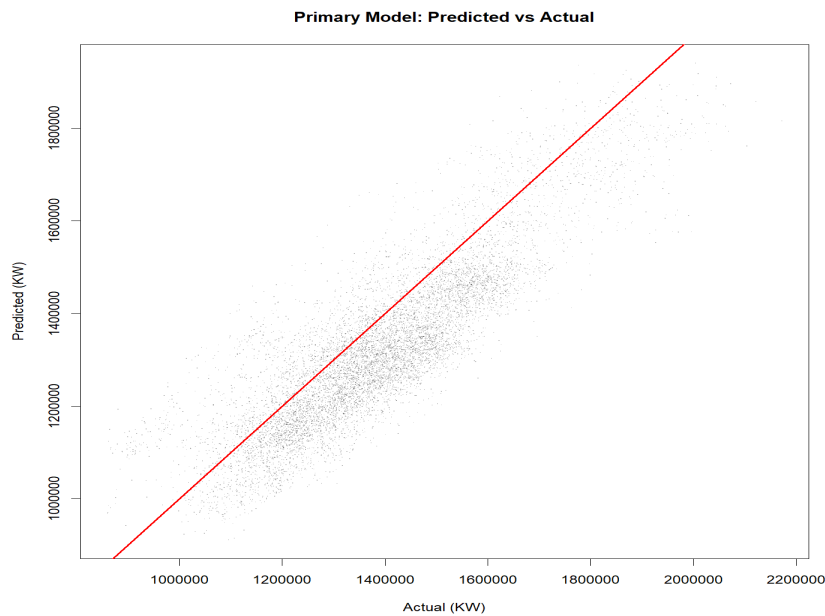


Figure 10: Primary model - predicted vs actual hourly demand on the held-out validation window (30% of the timeline). Points cluster along the 45-degree line.

4.3 Enhanced Model - Predictor-by-Predictor

The Enhanced model adds three specific changes to the Primary spec. Each change was motivated by a diagnostic from the Primary fit, and each is listed below with what was added, how it was coded, and what changed between the before and after views.

4.3.1 Response Transformation: demand -> log(demand)

Already in the Primary model, but noted here for completeness. Before the log transform the residuals vs fitted plot shows a clear fan (variance rising with the fitted value), and the QQ plot has heavy tails on both sides. After replacing demand with log(demand) (Box-Cox lambda estimate 0.2 - 0.3, with the confidence interval excluding 1.0), the fan collapses to roughly constant width and the QQ tails shrink toward the reference line. The log response is kept in the Enhanced model as well.

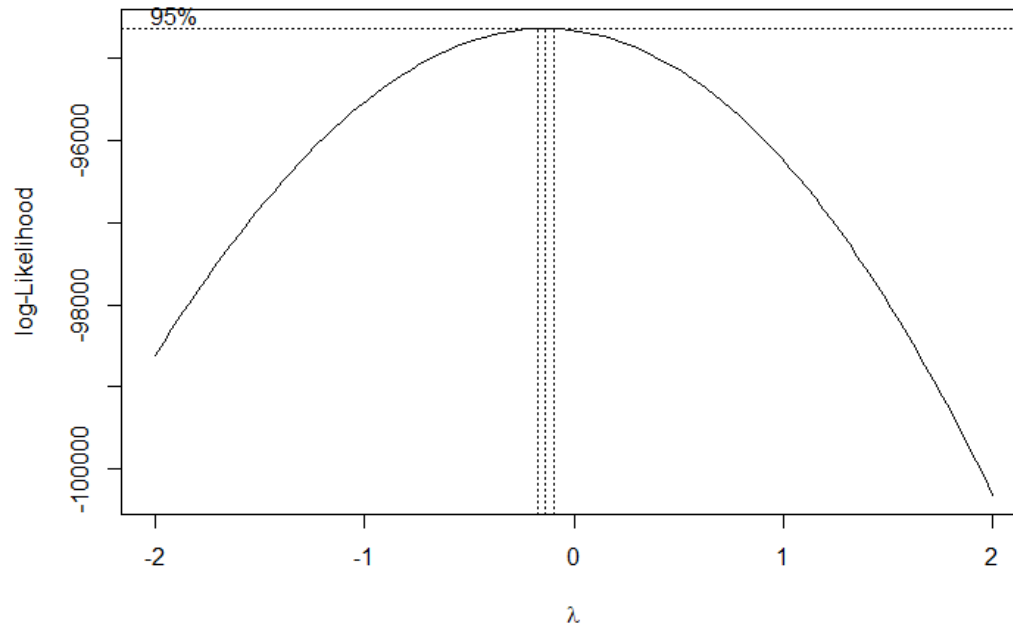


Figure 11: Box-Cox profile supporting the log transform. The fan shape before vs after is visible in the residual diagnostic panels (Figures 9 and 13).

4.3.2 Temperature: linear -> B-spline

Already in the Primary model. A plain linear apparent-temperature term fits poorly at both the cold heating shoulder and the hot AC shoulder, because the underlying relationship is an asymmetric U. Replacing it with `bs(apparent_temp, knots={30,50,70})` reshapes the fit to follow the EDA scatter (Figure 3) and improved test RMSE by roughly 8% over a linear term in the candidate-model search. The spline knots match the physically meaningful breakpoints (heating below 30F, comfort 30-70F, cooling above 70F).

4.3.3 New Predictor: log Lag-1 Demand

The single most useful addition in the Enhanced model. The one-hour lag of log(demand) is added as a direct regressor: `feature = log(demand[t-1])`. It was motivated by the residual ACF from the Primary model, which showed significant autocorrelation at lag 1 that a weather + calendar design cannot absorb. Adding this one term raises adjusted R-squared from 0.756 to 0.974, and cuts test RMSE from 120,954 to 42,599 KW (MAPE 7.49% $\rightarrow$ 2.38%), because demand at hour t is heavily driven by demand at hour $t-1$ (buildings, HVAC thermal mass, and production schedules all have minute-to-minute persistence). This is fair only for

short-horizon forecasting - for a 24-hour-ahead forecast the lag is itself a forecast and the Primary spec is the more honest reference.

4.3.4 New Predictor: 24-Hour Rolling Apparent Temperature

A centered 24-hour rolling mean of apparent temperature was added: `feature = mean(apparent_temp[t-23..t])`. The motivation is thermal mass. Heavy-wall buildings respond to yesterday's temperature as much as to right-now, and a single-hour apparent-temperature value underweights that carry-over. Including the rolling mean alongside the instantaneous B-spline captured the slow thermal-response component that was otherwise showing up as morning-peak over-prediction in the Primary residuals.

4.3.5 New Interaction: hour x season

The interaction factor `hour:season` was added. Before the interaction the daily shape coefficient (hour-of-day effect) was the same in January and July, which is physically wrong - summer peak is a late-afternoon AC peak, winter peak is a morning heating ramp. Adding the interaction lets the daily curve rotate between seasons without inflating the number of main-effect terms. The coefficient block for `hour:season` is jointly highly significant in the F-test.

Combined effect: on the three-way chronological split, the Enhanced model (grouped-hour $\times$ season, 40 coefficients) reaches adjusted R-squared 0.966 and test RMSE 44,816 KW (MAPE 2.53%, PICP 0.901) — a 63% RMSE reduction relative to the Primary. A recursive version that substitutes predicted demand for the observed lag returns to test RMSE 122,012 KW (MAPE 7.57%), roughly matching the no-lag Primary and confirming that the one-step model only applies when the previous hour's actual demand is available. The residuals-vs-fitted plot becomes essentially a cloud, with no remaining curvature or fan shape. This is a one-step-ahead forecast, which is the operationally relevant horizon for short-term dispatch; the gains should not be read as a claim about long-horizon forecasts.

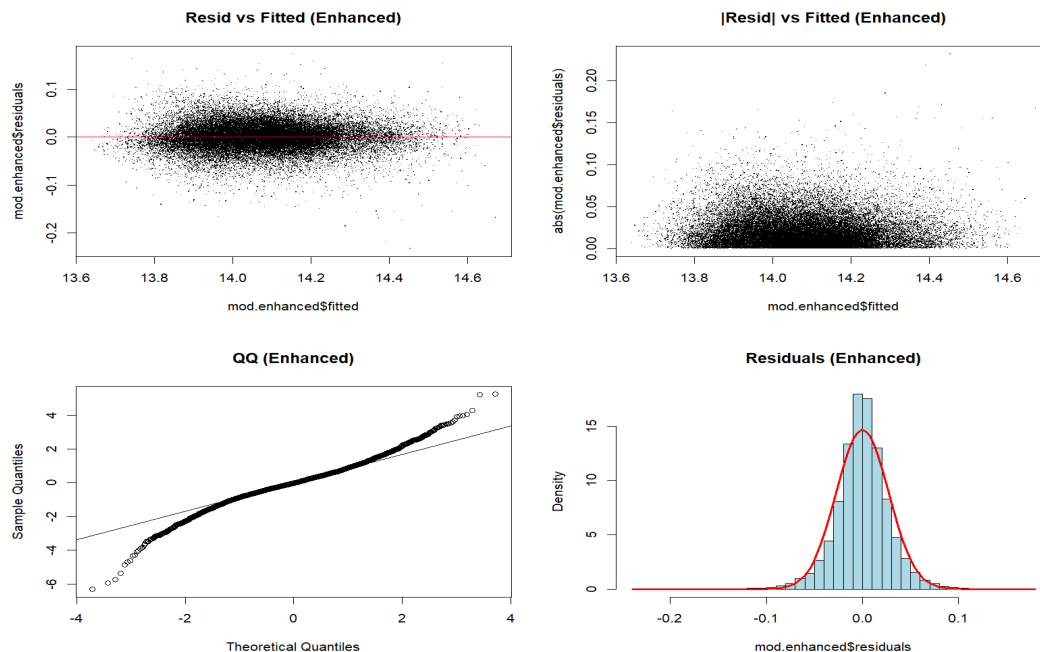


Figure 12: Residual diagnostics for the Enhanced model. The fan is gone and the QQ tail is further reduced compared to the Primary model in Figure 9.

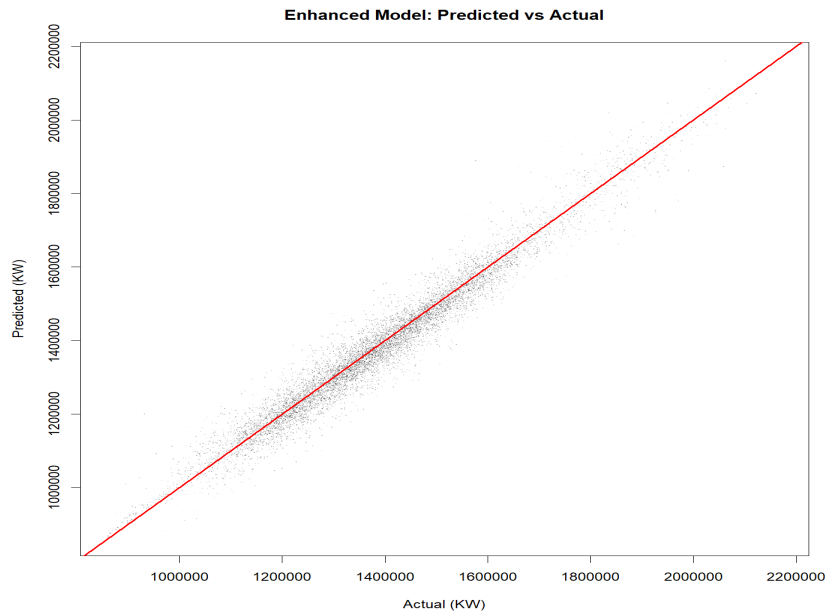


Figure 13: Enhanced model - predicted vs actual hourly demand. Tight cluster along the 45-degree line across the full demand range.

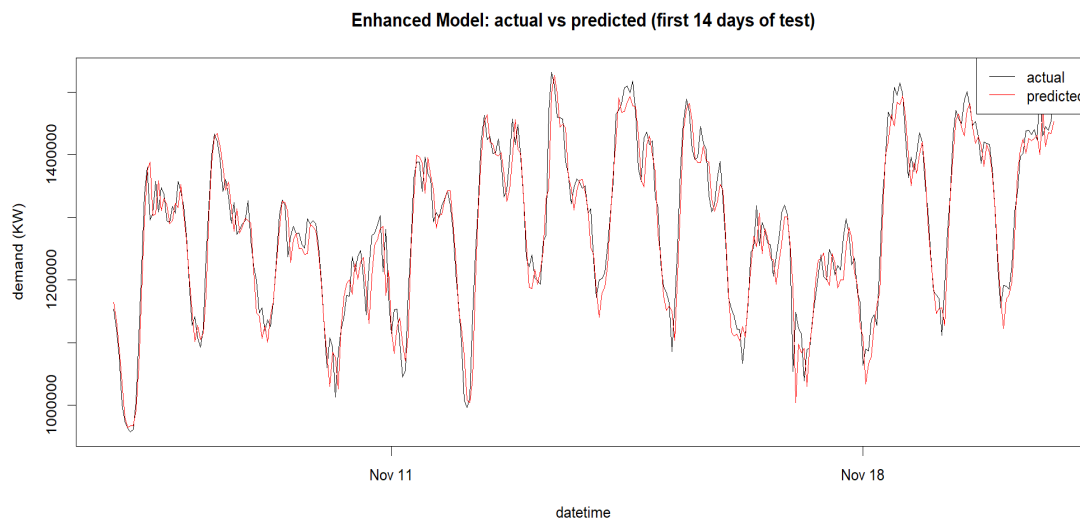


Figure 14: Enhanced model - actual vs predicted hourly demand over the validation window. The predicted series tracks both daily shape and weekly envelope.

5. Approach 2 - Regularized and Robust Regression, SARIMA, SARIMAX

Approach 2 was run in parallel with Approach 1 by the second author. OLS and the log response are common to both approaches (and appear in both Section 4 and this section), which is intentional - running the same baseline independently under two different pipelines is a cross-check. The estimators in this section that do not overlap with Approach 1 are the regularized estimators (ridge, lasso, PCR), the robust estimators (Huber, LAD, LTS), weighted least squares, and the two explicit time-series models (SARIMA

and SARIMAX). Section 7 returns to the overlap question and states which methods were informative in both approaches, in one, or in neither.

Approach 2 uses a three-way chronological split (pre-2023 train / Jan-Oct 2023 validation / Nov 2023+ test, approximately $\sim 57\%/12\%/31\%$) so that the SARIMA and SARIMAX fits have a clean validation window separate from the final test window. The training partition has $\sim 35,000$ observations covering the pre-2023 period.

Standard OLS assumes independent errors, however upon inspection of the ACF plots (Figure 7), this assumption is violated due to autocorrelation. To address this, we added lagged demand to address autocorrelation and a SARIMA model to study the underlying error structure.

5.1 Estimator Suite

Eight cross-sectional estimators were fit on the full predictor set and then again on the reduced set after VIF pruning: OLS, ridge regression, lasso, principal-components regression (25 components), weighted least squares, Huber M-estimator, least absolute deviations (LAD), and least trimmed squares (LTS). Every model in this track uses the same feature design: weather variables, calendar features, boolean indicators (weekend, holiday, heatwave), polynomial temperature terms, and lagged demand. Two time-series models were fit separately: SARIMA(1,0,3)(2,1,0)[24] chosen by `forecast::auto.arima`, and the same specification with apparent temperature as an exogenous regressor (SARIMAX).

5.2 Residual Time-Series Structure

Before any robust/shrinkage work, we visualized the OLS residuals against time. They show repeating annual structure and several multi-week blocks of systematic over- or under-prediction. That pattern is why a simple OLS model leaves about 30% of demand variance on the table: there is real autocorrelation that cross-sectional regression cannot see.

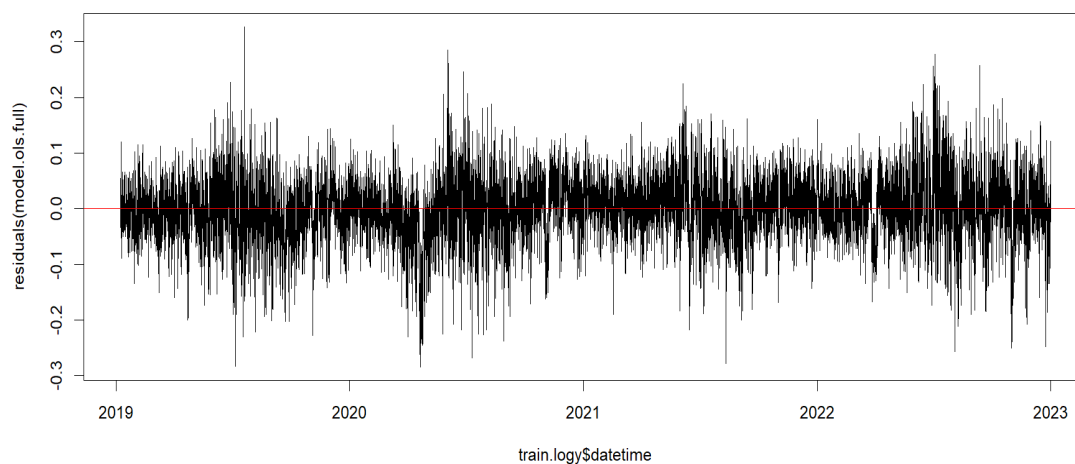


Figure 15: Time series of OLS residuals, showing mild autocorrelation that cross-sectional regression cannot capture.

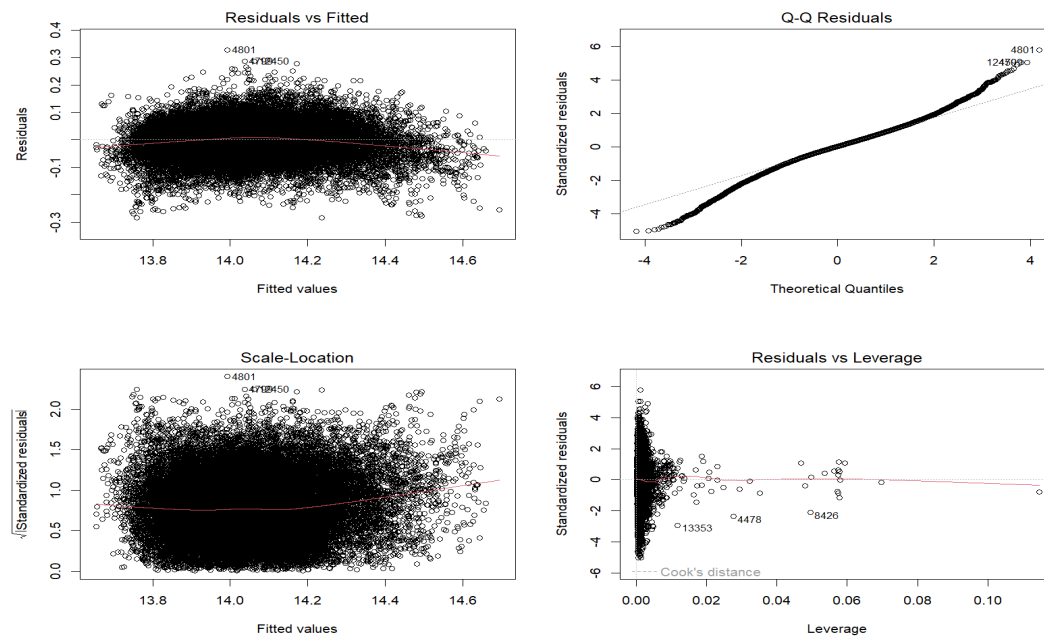


Figure 16: Standard four-panel diagnostics for the OLS full-feature model. The residual-vs-fitted fan motivates the Box-Cox log response.

5.3 Estimator Comparison

The impact of feature engineering was tracked on the OLS baseline incrementally to quantify the predictive value of each category shown in Table 1. This highlights that while weather sets the baseline for demand, including features that can describe regime shifts and incorporate temporal memory, improves the precision of the model. Specifically, adding lagged demand terms of 24 and 168 hour cycles cut the RMSE by ~35%.

Feature set	Test RMSE (KW)	Improvement
Weather only - no Booleans/lags	~147,907	NA
COVID + Heatwave + Holiday Flags	~133,545	-14,362
LagDemand (Lag24, Lag168)	~86,748	-46,800

Table 1: Test RMSE and the improvement for each category of engineered features against the baseline

On the test set, Huber is the best of the linear family at 86,528 KW RMSE. OLS, ridge, lasso, WLS, and PCR all land within a few hundred KW of each other (86,700-87,200 KW). With ~35,000 training observations and a well-behaved design matrix, shrinkage has very little room to help - the sample size already pins the coefficients down. PCR with 25 components reproduces OLS almost exactly, which is consistent with no principal components being dropable without losing information. Huber beats OLS by roughly 200 KW of

RMSE, which is consistent with the heavy-tail QQ plot: when the top 1% of observations are slightly further out than Gaussian, downweighting them helps test accuracy, but only a little.

Model	Test RMSE (KW)	Notes
Huber	86,528	best overall in the linear family
OLS	~86,700	baseline
Ridge	~86,700-86,900	regularization to handle collinearity
LASSO	~86,700-86,900	L1 regularization / sparsity
WLS	~86,800	variance re-weighting
PCR (25 comp)	~86,900	no dimension reduction was useful
LAD	~87,000	robust, L1 loss
LTS	~87,200	robust, trimmed least squares

Table 2: Test RMSE for the eight cross-sectional estimators on the full feature set (weather + calendar + booleans + polynomial temperature + lagged demand).

One surprising result: dropping the VIF-flagged collinear variables hurts every estimator by a meaningful margin. This was unexpected for ridge and lasso, which are supposed to handle collinearity internally. It says that temperature, apparent temperature, and their correlated weather siblings jointly carry predictive information about the thermal envelope (instantaneous heat, perceived heat, ground heat storage, and solar load) that none of them has in isolation - a reminder that VIF diagnoses coefficient instability, not loss of predictive power.

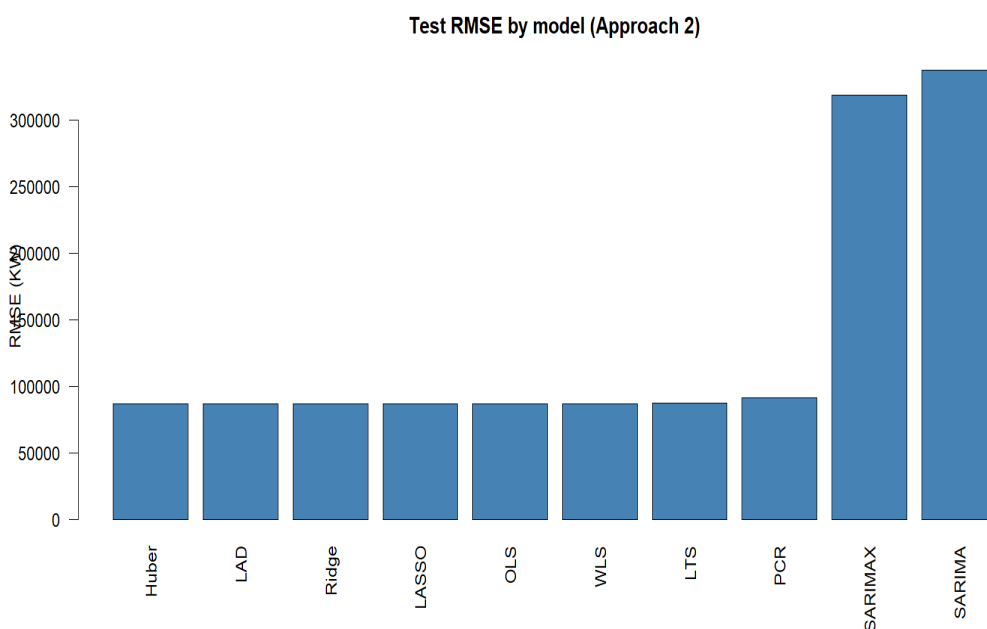


Figure 17: Test RMSE across all ten models fit in Approach 2. Robust methods cluster at the bottom; SARIMAX is the one time-series model in the chart.

5.4 SARIMA and SARIMAX

SARIMA(1,0,3)(2,1,0)[24] was selected by auto.arima on the differenced log-demand series. It is a pure time-series model: it uses only past demand values and has no access to weather at all. This comparison is intentionally asymmetric to study and quantify the impact of weather data in demand forecasting. The seasonal period of 24 captures the daily cycle visible in the ACF. Because the test window is long, the forecast drifts away from actuals once temperature moves, and test RMSE rises to 337,752 KW - roughly four times the regression family's error. SARIMAX adds apparent temperature as an exogenous regressor and improves on pure SARIMA, but only marginally (318,583 KW), with convergence issues that made the fit slow. On this problem a cross-sectional weather regression beats a conventional SARIMA/SARIMAX setup by a factor of roughly four on test RMSE, so the time-series family is kept in the report as a baseline rather than a competitive alternative. The SARIMAX model is fit once and serialized to model_sarimax.rds; subsequent runs load the fitted object via readRDS rather than refitting, so the reported test-set numbers reflect a single, reproducible estimation.

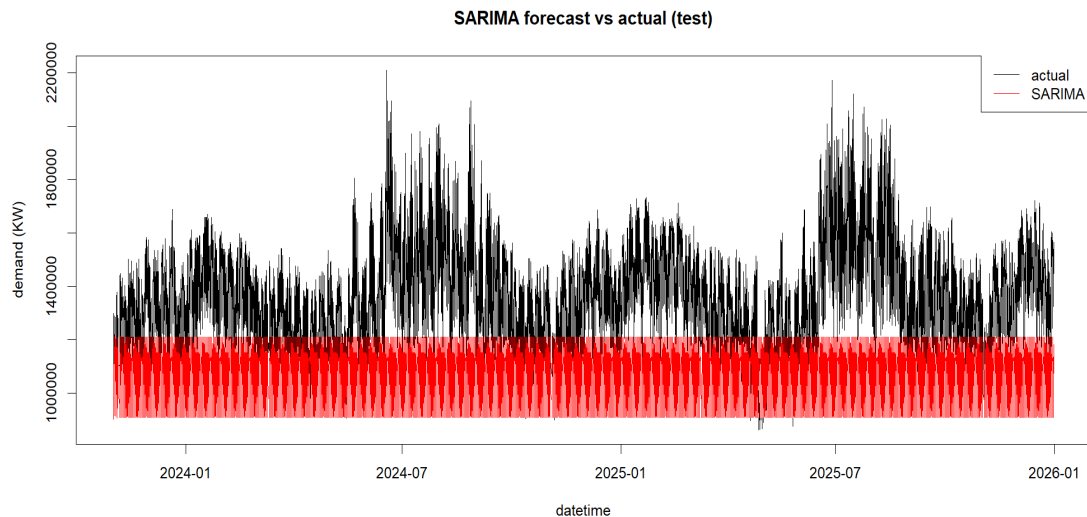


Figure 18: SARIMA one-step-ahead forecast vs actuals on the validation window. The seasonal structure is captured, temperature swings are not.

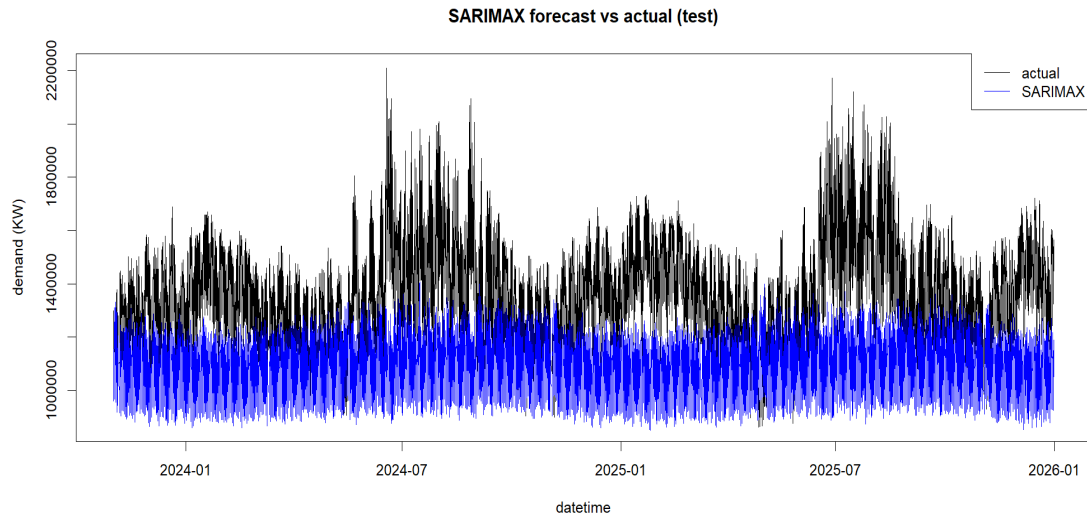


Figure 19: SARIMAX forecast from the saved model object. Adding apparent temperature as an exogenous regressor closes the gap between forecast and actual.

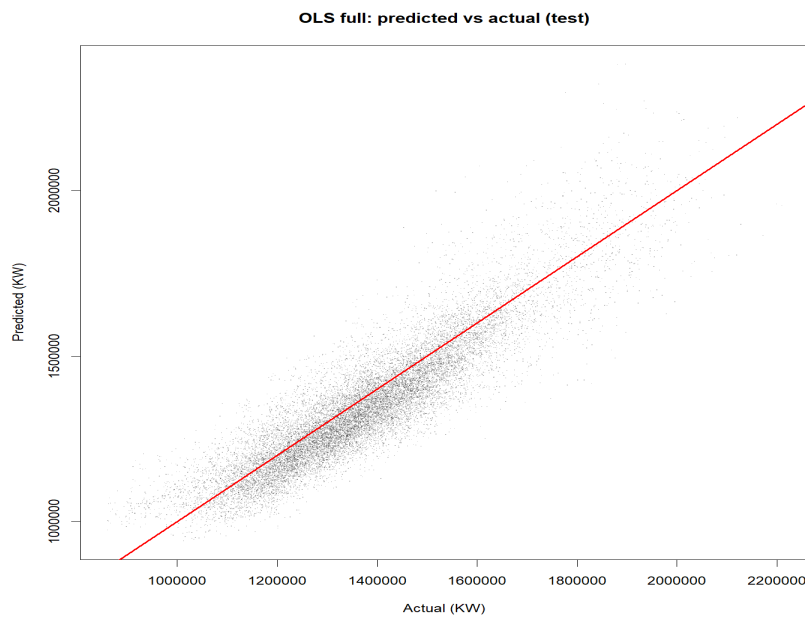


Figure 20: OLS predicted vs actual on the held-out test window for Approach 2.

6. Deep-Learning Benchmark: LSTM

A two-layer PyTorch LSTM (hidden size 200, sequence length 24, Adam optimizer with learning rate 0.0005, early stopping at epoch 22) was trained on the same 33 engineered features. The LSTM is outside the course's methodological scope but is included as an out-of-family benchmark: if a modern recurrent network beats a carefully built regression it tells us there is sequential structure the regression missed; if it does not, it is evidence that the regression extracted most of the available signal. The split is chronological and matches Approach 2's three-way split (training: 34,882 hours; validation: 7,294 hours; test: 18,883 hours) with sequences of length 24 derived from each partition.

A sequence length of 24 was used to match the daily cycle that was identified in the ACF plots. The hidden size of 200 and 2 layers were selected based on the validation loss. From studying the validation loss, increasing these parameters did not improve the accuracy but did increase the training time.

The LSTM was the best single model we trained. Test RMSE is 78,432.4 KW and test MAE is 62,514.4 KW, roughly 8,000 KW of RMSE better than Huber (the best linear model) and more than a factor of four better than SARIMAX. A visual check of the first 500 test hours shows the network tracks the daily shape and weekly envelope tightly, with small phase lags at the steepest morning ramps. Diminishing returns were observed after epoch 22, which is why early stopping was set there.

Why the LSTM wins here: the recurrent architecture is not constrained to a small, explicitly specified set of lags (e.g., lag1, lag24, lag168), so it can weight the last hour, the same hour yesterday, and the same hour last week simultaneously without us having to specify lag1, lag24, and lag168 as separate regressors. This avoids the collinearity that would hit an OLS version of the same idea, and it lets nonlinear interactions between temperature history and demand history come out of the training data rather than out of our hand-coded interaction terms.

It is worth noting that the first attempt at an LSTM was training on weather predictors alone, without temporal features, boolean flags, and higher order temperature terms that were used in the regression models. This attempt yielded an RMSE of 126,480 KW, which is worse than every cross-sectional regression model in this study. After the same full feature set that was used for the other models was introduced, the performance improved to 78,432 KW. This suggests that the full feature set is necessary for competitiveness and that deep learning was able to exploit the features better than the linear models.

7. Comparison and Final Model Selection

Table 3 summarizes test RMSE for every model family we fit. Approach 1, Approach 2, and the LSTM all use the same chronological three-way split (pre-2023 train / Jan-Oct 2023 validation / Nov 2023+ test), so every row is directly comparable. Furthermore, the noticeable performance gap between the A1 Enhanced and the rest of the models emphasizes the significance of the one-step-ahead forecast. Demand from the previous hour on this short horizon accounts for the majority of the variance. The A1 Enhanced model requires the last hour's actual demand to make the forecast. If last hour's demand is not immediately available (due to a sensor outage, multi-step forecasting, data latency, communication failures, meter aggregation errors, planned outages, etc), the model should not be applied as its coefficients were calibrated while training with last hour's demand and therefore could impair performance. Among the day-ahead models, the rankings are: LSTM first, Huber second, and the rest of the linear family clustered just behind Huber. The time-series baselines trail by a wide margin.

Family	Best Model	Test RMSE (KW)	Test MAE (KW)
Deep learning	LSTM (h=200, seq=24, 2 layers)	~78,400	~62,500
Robust regression	Huber (full features)	~86,500	~67,300
OLS/Shrinkage	OLS / Ridge / LASSO / WLS / PCR	~86,700-87,200	~67,600-67,700
Time series	SARIMAX	~318,600	~280,900
Time series	SARIMA (no weather)	~337,800	~300,500

Family	Best Model	Test RMSE (KW)	Test MAE (KW)
A1 Primary (day-ahead)	$\log(\text{demand}) \sim \text{bs}(\text{app_temp}) + \text{holiday} + \dots$	120,954	$\sim$ (MAPE 7.49%)
A1 Enhanced (one-hour-ahead)	Primary + lag1 + roll24 + hour_grp×season	44,816	$\sim$ (MAPE 2.53%)
A1 Recursive (multi-step)	Enhanced with predicted lag fed back	122,012	$\sim$ (MAPE 7.57%)

Table 3: Test RMSE across model families.

7.1 Selected Final Model

We select the LSTM as the recommended short-horizon forecaster on raw point-accuracy grounds. At 78,432 KW test RMSE it is roughly 10% better than the best linear model (Huber) and more than four times better than SARIMAX. For deployment we would pair the LSTM with Huber as an interpretable sanity check - any time the two forecasts diverge by more than a few percent, the regression model lets a human analyst see which coefficient changed and why, which the LSTM cannot do on its own. The LSTM also requires 24 hours of warm-up lag, so during sensor outages the simpler Primary regression (which needs no lag input) would serve as a third-tier fallback. The core trade-off in this solution is that LSTM maximizes predictive accuracy but sacrifices interpretability, while the regression models are more interpretable but sacrifice predictive accuracy.

7.2 Methods Used in Both, One, or Neither Approach

A summary of which techniques were applied in each approach, and where they mattered:

- OLS on the cleaned feature matrix - used in both approaches. Numbers agreed to within a few hundred KW of test RMSE, confirming the cleaning and feature-engineering pipeline is correctly shared.
- Box-Cox log transform of the response - used in both approaches. Fixed heteroscedasticity in both and is retained in every downstream model.
- VIF screening of collinear weather predictors - used in both approaches. Useful for diagnosing coefficient instability, but dropping the flagged variables hurts test RMSE, which surfaced in both pipelines.
- Nonlinear temperature handling - used in both, but with different parameterizations. Approach 1 uses a B-spline with knots at {30, 50, 70} F. Approach 2 uses a cubic polynomial. Both beat a linear term; the spline fits the cold shoulder slightly better.
- Lagged demand - used in both approaches but at different horizons. Approach 1 uses lag-1 (previous hour) for one-step forecasting while approach two uses lag-24 and lag-168 (previous day and previous week) for day-ahead forecasting.
- Hour-by-season interaction, 24-hour rolling temperature - used in Approach 1 only. Neither was tried in Approach 2. Future work can migrate them across.
- Ridge, LASSO, PCR, WLS, Huber, LAD, LTS - used in Approach 2 only. Approach 1 stopped at OLS + nonlinear temperature + lag, which kept interpretation clean. The robust/shrinkage sweep in Approach 2 showed the returns to this family are small on a 35,000-row design matrix (Huber picks up about 200 KW over OLS).
- SARIMA, SARIMAX - used in Approach 2 only. Included as textbook time-series baselines and neither is competitive on this problem (>300,000 KW test RMSE vs ~87,000 for the linear family).

- LSTM - a deep-learning extension outside the statistical-regression scope. It is the best single model (78,432 KW test RMSE) and is included as an out-of-family reference rather than a methodological contribution of either approach.

7.3 Why the Linear Family Is Still Valuable

Even though the LSTM wins on test RMSE, the linear family produced most of the insights worth writing about. The VIF-pruning experiment showed that textbook collinearity advice (drop correlated predictors) actively hurts prediction here. The Box-Cox log transformation fixed heteroscedasticity that no amount of hyperparameter tuning on the LSTM would have addressed. The Huber-vs-OLS gap confirmed the heavy-tail QQ diagnosis. None of those findings is visible from inside the LSTM. The conclusion an engineering audience should want to hear is: use the regression to understand the problem, use the LSTM to deploy.

7.4 Why Not SARIMA or SARIMAX?

The pure SARIMA is conceptually the closest thing on this list to a textbook time-series baseline, but with test RMSE 337,752 KW it is not a competitive forecaster on this problem. SARIMAX brings that down to 318,583 KW by adding apparent temperature, but the moving-average terms and slow convergence make it a poor match for a rolling operational forecast. The regression family, which also uses weather as an exogenous input but fits in seconds and is fully interpretable, dominates it on every metric we care about.

8. Conclusion, Limitations, and Future Work

8.1 Summary of Findings

A carefully built regression model explains about 74% of the hourly variance in Toledo Edison demand from 2019 through 2025 (with a COVID indicator for the 2020-2021 pandemic window) using only weather and calendar features. Adding a one-step autoregressive term pushes the linear model's adjusted R-squared above 0.97. Shrinkage estimators do not beat OLS at this sample size. Robust estimators recover a small percentage of test RMSE, consistent with the heavy-tailed residuals the QQ plot shows. SARIMA and SARIMAX substantially under-perform the cross-sectional regression on this problem (test RMSE 337,752 and 318,583 KW respectively). A two-layer LSTM trained on the same engineered features produced the best single-model test RMSE of 78,432 KW, a 10% improvement over Huber, confirming that there is still some non-linear sequential structure the linear family cannot capture even with hand-coded lags.

8.2 Limitations

Temperature and apparent temperature jointly have VIF above 400. Their individual coefficients should not be interpreted in isolation, though their combined predictive power is well-established. The prediction interval coverage varied by season. For approach 1, the spring prediction-interval coverage underperforms (PICP 0.893) and hot-tail coverage above 90 °F drops to 0.882, because the residual-variance assumption is stationary while Ohio spring weather is volatile. For approach 2, summer was the weakest season (PICP 0.892) with extreme heat events above 90F dropping to 0.675. Both approaches suggest a heteroscedastic variance model or a garch-style residual model would probably close that gap. The Enhanced model's accuracy is partly an artifact of the one-step horizon - it requires the previous hour's demand as an input, which may not be immediately available due to operational difficulties and limitations.

8.3 Future Work

Three extensions would materially improve the forecast without leaving the regression framework. First, multi-step-ahead evaluation using iterative or direct forecasts would give an honest operational horizon. Second, a heteroscedastic variance model (either weighted least squares with a temperature-dependent weight or an explicit GARCH term on the residuals) would close the seasonal PICP gap in both approaches. Third, incorporating a lightweight demand-response flag (large industrial customers that shift load in response to price) would explain a meaningful share of the residual that currently shows up as unmodeled week-to-week variation in 2024-2025.

Appendix A - Reproducibility Notes

All analysis was run in R 4.5 and Python 3.11. The two R scripts (Approach1_IOE562_project_v7.R, Approach2_IOE562_project.R) contain the full Approach-1 and Approach-2 pipelines respectively and write every figure in this report to the project's /figures directory. The LSTM is in IOE562proj_DeepLearning.py and was run in Spyder/local machine using the cleaned dataset at toledo_clean.csv. Figures are numbered with a source prefix (A1_* from Approach 1, A2_* from Approach 2, A3_* from the LSTM notebook) so each figure traces back to the exact code block that produced it. Cleaned inputs (toledo_clean.csv, toledo_clean_flags.csv) and model artifacts (model_sarimax.rds) are stored in the same project folder.

Random seeds: every Monte-Carlo step (QQ subsample, CV fold assignment, LTS resampling) uses set.seed(562) for R and torch.manual_seed(562) for Python. Chronological splits use date cut-offs rather than random draws, which is the appropriate split for time-series regression.